

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** Xylene - S

|                                |                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>R40133</b>                                                                                        |
| <b>Address</b>                 | ThermoFisher Scientific Australia Pty Ltd<br>5 Caribbean Drive, Scoresby<br>VICTORIA 3179, Australia |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>03 9757 4559 or +613 9757 4559</b>                                            |
| <b>Telephone / Fax Numbers</b> | Tel: 1300 735 292<br>Fax: 1800 067 639                                                               |
| <b>E-mail address</b>          | ANZinfo@thermofisher.com                                                                             |

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

Flammable liquids

Category 3

#### Health hazards

Aspiration Toxicity  
Skin Corrosion/Irritation  
Skin Sensitization

Category 1  
Category 2  
Category 1

#### Environmental hazards

Acute aquatic toxicity  
Chronic aquatic toxicity

Category 1  
Category 1

#### Label Elements



Flame



Health Hazard



Exclamation Mark



Environment

**Signal Word****Danger****Hazard Statements**

H226 - Flammable liquid and vapor  
 H304 - May be fatal if swallowed and enters airways  
 H315 - Causes skin irritation  
 H317 - May cause an allergic skin reaction  
 H410 - Very toxic to aquatic life with long lasting effects

**Precautionary Statements**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
 P233 - Keep container tightly closed  
 P240 - Ground and bond container and receiving equipment  
 P242 - Use non-sparking tools  
 P243 - Take action to prevent static discharges  
 P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
 P264 - Wash face, hands and any exposed skin thoroughly after handling  
 P272 - Contaminated work clothing should not be allowed out of the workplace  
 P280 - Wear protective gloves/protective clothing/eye protection/face protection  
 P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor  
 P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
 P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
 P331 - Do NOT induce vomiting  
 P333 + P313 - If skin irritation or rash occurs: Get medical advice/attention  
 P337 + P313 - If eye irritation persists: Get medical advice/attention  
 P363 - Wash contaminated clothing before reuse  
 P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
 P403 + P235 - Store in a well-ventilated place. Keep cool  
 P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

This product does not contain any known or suspected endocrine disruptors

## Section 3 - Composition and Information on Ingredients

| Component                | CAS No     | Weight % |
|--------------------------|------------|----------|
| D-Limonene               | 5989-27-5  | >98.0    |
| Butylated hydroxyanisole | 25013-16-5 | <0.024   |

## Section 4 - First Aid Measures

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur. Risk of serious damage to the lungs (by aspiration).

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Do NOT induce vomiting. Call

|                                            |                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.                                                                                                                                                                                                                             |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                                                                                                                                                                                                     |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                                                                                                                       |
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                                                                                                                                                                                                                                |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                      |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                                  |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. May cause allergic skin reaction. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Notes to Physician</b>                  | Treat symptomatically. Symptoms may be delayed.                                                                                                                                                                                                                                                                                       |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

Do not use a solid water stream as it may scatter and spread fire.

### Hazardous Decomposition Products

Carbon oxides, Thermal decomposition can lead to release of irritating gases and vapors.

### Specific Hazards Arising from the Chemical

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Do not allow run-off from fire-fighting to enter drains or water courses.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental Precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. See Section 12 for additional Ecological Information. Avoid release to the environment. Collect spillage.

### Methods for Containment and Clean Up

#### Clean-up methods - small spillage

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Take precautionary measures against static discharges. Use spark-proof tools and explosion-proof equipment.

#### Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

#### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

#### Precautions for Safe Handling

Ensure adequate ventilation. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Use spark-proof tools and explosion-proof equipment. Take precautionary measures against static discharges.

#### Conditions for Safe Storage, Including any Incompatibilities

Keep away from heat, sparks and flame. Keep tightly closed in a dry and cool place. Flammables area. Keep container tightly closed in a dry and well-ventilated place.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

#### Exposure limits

DE - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component                | Australia | New Zealand WEL | ACGIH TLV | The United Kingdom | Germany                                                                                                                                                                                                                                                      |
|--------------------------|-----------|-----------------|-----------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D-Limonene               |           |                 |           |                    | TWA: 5 ppm (8 Stunden). AGW - exposure factor 4<br>TWA: 28 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 4<br>TWA: 5 ppm (8 Stunden). MAK<br>TWA: 28 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 20 ppm<br>Höhepunkt: 112 mg/m <sup>3</sup> Haut |
| Butylated hydroxyanisole |           |                 |           |                    | TWA: 20 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 1<br>TWA: 20 mg/m <sup>3</sup> (8 Stunden). MAK can occur as vapor and aerosol at the same time<br>Höhepunkt: 20 mg/m <sup>3</sup>                                                              |

#### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

#### Exposure Controls

##### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to

control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Viton (R)      | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Skin and body protection

Long sleeved clothing

#### Respiratory Protection

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

#### Recommended Filter type:

Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)

#### Recommended half mask:-

Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent) When RPE is used a face piece Fit Test should be conducted

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

#### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                         |                                    |                                   |
|-----------------------------------------|------------------------------------|-----------------------------------|
| Appearance                              | Clear - Colourless                 |                                   |
| Physical State                          | Liquid                             |                                   |
| Odor                                    | Citrus                             |                                   |
| Odor Threshold                          | No data available                  |                                   |
| pH                                      | No information available           |                                   |
| Melting Point/Range                     | No data available                  |                                   |
| Softening Point                         | No data available                  |                                   |
| Boiling Point/Range                     | 175 °C / 347 °F                    |                                   |
| Flash Point                             | 48 °C / 118.4 °F                   |                                   |
| Evaporation Rate                        | No data available                  | Method - No information available |
| Flammability (solid,gas)                | Not applicable                     | Liquid                            |
| Explosion Limits                        | Lower 0.7 vol %<br>Upper 6.1 vol % |                                   |
| Vapor Pressure                          | No data available                  |                                   |
| Vapor Density                           | No data available                  | (Air = 1.0)                       |
| Specific Gravity / Density              | No data available                  |                                   |
| Bulk Density                            | Not applicable                     | Liquid                            |
| Water Solubility                        | No information available           |                                   |
| Solubility in other solvents            | No information available           |                                   |
| Partition Coefficient (n-octanol/water) |                                    |                                   |
| Component                               | log Pow                            |                                   |
| D-Limonene                              | 4.38                               |                                   |

|                           |                          |                                        |
|---------------------------|--------------------------|----------------------------------------|
| Butylated hydroxyanisole  | 3.335                    |                                        |
| Autoignition Temperature  | 255 °C / 491 °F          |                                        |
| Decomposition Temperature | No data available        |                                        |
| Viscosity                 | No data available        |                                        |
| Explosive Properties      |                          | explosive air/vapour mixtures possible |
| Oxidizing Properties      | No information available |                                        |

Other information

## Section 10 - Stability and Reactivity

|                                  |                                                                                                             |
|----------------------------------|-------------------------------------------------------------------------------------------------------------|
| Reactivity                       | None known, based on information available                                                                  |
| Stability                        | Stable under recommended storage conditions.                                                                |
| Conditions to Avoid              | Heat, flames and sparks, Exposure to air, Keep away from open flames, hot surfaces and sources of ignition. |
| Incompatible Materials           | None known.                                                                                                 |
| Hazardous Decomposition Products | Carbon oxides. Thermal decomposition can lead to release of irritating gases and vapors.                    |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                    |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

|                     |                                                                                          |
|---------------------|------------------------------------------------------------------------------------------|
| Product Information | Product does not present an acute toxicity hazard based on known or supplied information |
|---------------------|------------------------------------------------------------------------------------------|

**(a) acute toxicity;****Oral**

Based on ATE data, the classification criteria are not met

**Dermal**

Based on ATE data, the classification criteria are not met

**Inhalation**

Based on available data, the classification criteria are not met

| Component                | LD50 Oral                 | LD50 Dermal               | LC50 Inhalation |
|--------------------------|---------------------------|---------------------------|-----------------|
| D-Limonene               | LD50 = 5200 mg/kg ( Rat ) | LD50 > 5 g/kg ( Rabbit )  |                 |
| Butylated hydroxyanisole | LD50 = 2 g/kg ( Rat )     | LD50 > 2000 mg/kg ( Rat ) |                 |

**(b) skin corrosion/irritation;**

Category 2

**(c) serious eye damage/irritation;**

Based on available data, the classification criteria are not met

**(d) respiratory or skin sensitization;****Respiratory**

Based on available data, the classification criteria are not met

**Skin**

Category 1

**Sensitization**

No information available

**(e) germ cell mutagenicity;**

Based on available data, the classification criteria are not met

**(f) carcinogenicity;**

Based on available data, the classification criteria are not met

This product contains one or more substances which are classified by IARC as carcinogenic to humans (Group I), probably carcinogenic to humans (Group 2A) or possibly

carcinogenic to humans (Group 2B) The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component                | Australia | New Zealand | New South Wales | Western Australia | IARC     | EU | UK | Germany |
|--------------------------|-----------|-------------|-----------------|-------------------|----------|----|----|---------|
| Butylated hydroxyanisole |           |             |                 |                   | Group 2B |    |    |         |

(g) reproductive toxicity; Based on available data, the classification criteria are not met

(h) STOT-single exposure; Based on available data, the classification criteria are not met

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Target Organs None known.

(j) aspiration hazard; Category 1

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

## Section 12 - Ecological Information

**Ecotoxicity effects** Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component  | Freshwater Fish                                                                                                | Water Flea | Freshwater Algae | Microtox |
|------------|----------------------------------------------------------------------------------------------------------------|------------|------------------|----------|
| D-Limonene | LC50: = 35 mg/L, 96h (Oncorhynchus mykiss)<br>LC50: 0.619 - 0.796 mg/L, 96h flow-through (Pimephales promelas) |            |                  |          |

**Persistence and Degradability** No information available  
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.  
**Bioaccumulative Potential** No information available

| Component                | log Pow | Bioconcentration factor (BCF) |
|--------------------------|---------|-------------------------------|
| D-Limonene               | 4.38    | No data available             |
| Butylated hydroxyanisole | 3.335   | No data available             |

**Mobility** No information available.

### Endocrine Disruptor Information

| Component                | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances | Japan - Endocrine Disruptor Information |
|--------------------------|------------------------------------------|--------------------------------------------------|-----------------------------------------|
| Butylated hydroxyanisole | Group III Chemical                       |                                                  |                                         |

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

**Waste from Residues/Unused Products** Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information**

Chemical wastes should be disposed through a licensed commercial waste collection service. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not let this chemical enter the environment. Do not empty into drains.

## Section 14 - Transport Information

**IMDG/IMO**

|                      |                              |
|----------------------|------------------------------|
| UN-No                | UN2319                       |
| Proper Shipping Name | TERPENE HYDROCARBONS, N.O.S. |
| Hazard Class         | 3                            |
| Packing Group        | III                          |

**ADG**

|                      |                              |
|----------------------|------------------------------|
| UN-No                | UN2319                       |
| Proper Shipping Name | TERPENE HYDROCARBONS, N.O.S. |
| Hazard Class         | 3                            |
| Packing Group        | III                          |

**IATA**

|                      |                              |
|----------------------|------------------------------|
| UN-No                | UN2319                       |
| Proper Shipping Name | TERPENE HYDROCARBONS, N.O.S. |
| Hazard Class         | 3                            |
| Packing Group        | III                          |

|                       |                                                                                                          |
|-----------------------|----------------------------------------------------------------------------------------------------------|
| Environmental hazards | Dangerous for the environment<br>Product is a marine pollutant according to the criteria set by IMDG/IMO |
|-----------------------|----------------------------------------------------------------------------------------------------------|

|                     |                                 |
|---------------------|---------------------------------|
| Special Precautions | No special precautions required |
|---------------------|---------------------------------|

|                        |            |
|------------------------|------------|
| Additional information | None known |
|------------------------|------------|

## Section 15 - Regulatory Information

**Safety, health and environmental regulations/legislation specific for the substance or mixture****National Regulations**      **Australia**

See section 8 for national exposure control parameters.

**Standard for the Uniform Scheduling of Medicines and Poisons**

No poison schedule number allocated.

**Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component              | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information                                                                                                                                                                                                       |
|------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D-Limonene - 5989-27-5 | Present                                                     | Specific information requirement: Obligations to provide information apply. You must tell us within 28 days if the circumstances of your importation or manufacture (introduction) are different to those in our assessment. |



|                                       |         |   |
|---------------------------------------|---------|---|
| Butylated hydroxyanisole - 25013-16-5 | Present | - |
|---------------------------------------|---------|---|

**Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

**Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory** Not applicable

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

**International Inventories**

| Component                | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|--------------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| D-Limonene               | X    | X     | 227-813-5 | -      | X    | X   | -    | X     | X    | X    | X     | KE-24397 |
| Butylated hydroxyanisole | X    | X     | 246-563-8 | -      | X    | X   | -    | X     | X    | X    | X     | KE-11392 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**

Not applicable.

| Component                | CAS No     | OECD HPV       | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|--------------------------|------------|----------------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| D-Limonene               | 5989-27-5  | Listed         | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |
| Butylated hydroxyanisole | 25013-16-5 | Not applicable | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

**Authorisation/Restrictions according to EU REACH**

| Component  | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| D-Limonene | -                                                                   | Use restricted. See item 75.                                                  | -                                                                                                     |

|  |                                    |  |
|--|------------------------------------|--|
|  | (see link for restriction details) |  |
|--|------------------------------------|--|

<https://echa.europa.eu/substances-restricted-under-reach>

## Section 16 - Other Information

### Legend

|                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>AICS</b> - Australian Inventory of Chemical Substances                                            | <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                                            |
| <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                      | <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances |
| <b>DSL/NDL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                      | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                                                  |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                     | <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                            | <b>CAS</b> - Chemical Abstracts Service                                                                                      |
| <b>TWA</b> - Time Weighted Average                                                                   | <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists                                                     |
| <b>IARC</b> - International Agency for Research on Cancer                                            | Predicted No Effect Concentration (PNEC)                                                                                     |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code                            |
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships                  |                                                                                                                              |
| <b>NZS 5433:2020</b> - Transport of Dangerous Goods on Land                                          | <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         |
| <b>LD50</b> - Lethal Dose 50%                                                                        | <b>LC50</b> - Lethal Concentration 50%                                                                                       |
| <b>EC50</b> - Effective Concentration 50%                                                            | <b>ATE</b> - Acute Toxicity Estimate                                                                                         |
| <b>WEL</b> - Workplace Exposure Limit                                                                | <b>RPE</b> - Respiratory Protective Equipment                                                                                |
| <b>DNEL</b> - Derived No Effect Level                                                                | <b>NOEC</b> - No Observed Effect Concentration                                                                               |
| <b>POW</b> - Partition coefficient Octanol:Water                                                     | <b>BCF</b> - Bioconcentration factor                                                                                         |
| <b>vPvB</b> - very Persistent, very Bioaccumulative                                                  | <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              |
| <b>VOC</b> - (Volatile Organic Compound)                                                             |                                                                                                                              |

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

|                         |                       |
|-------------------------|-----------------------|
| <b>Revision Date</b>    | 30-Jun-2023           |
| <b>Revision Summary</b> | Update to CLP Format. |

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet